FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAIN PROJECT LOG BOOK

ACADEMIC YEAR 2025-26

Focus on Excellence

IntelShare AI: AI For Everyone Everywhere

GUIDE: Ms. Chethna Joy

Week	Date	Aman Mohamed K	Cyril Pius	Azim Palakkal	Anirudh Anilkumar	Remarks
Week 1	01/08/2025 – 16/08/2025	Studied survey papers on user-friendly AI interfaces and no-code ML platforms. Identified interaction patterns for non-technical users.	Reviewed initial papers to identify common themes. Studied social impact of accessible AI in education and assistive technology.	Explored papers on model interchange formats (ONNX) and collaborative AI platforms. Summarized sharing mechanisms.	Investigated real-time feedback mechanisms in ML. Studied hardware APIs for camera and microphone integration.	
Week 2	21/08/2025 – 02/09/2025	Researched lightweight vision models suitable for edge devices (MobileNet, MediaPipe). Analyzed trade-offs between accuracy and latency.	Studied ethical considerations in user-generated AI data and privacy. Explored consent-based interaction models.	Analyzed papers on incremental and online learning approaches. Studied confidence handling in ML systems.	Consolidated research findings. Drafted initial problem statement and project motivation.	
Week 3	04/09/2025 – 09/09/2025	Designed initial vision pipeline concept including embedding extraction and similarity matching.	Proposed voice-based interaction flow and confirmation mechanisms.	Defined knowledge unit structure and incremental update logic.	Drafted overall system architecture and high-level workflow.	
Week 4	10/09/2025 – 17/09/2025	Reviewed and refined system architecture diagrams with focus on vision boundaries.	Reviewed interaction flow for human-in-the-loop control.	Refined learning workflow to avoid dataset-based training.	Verified presentation content and aligned architecture with project scope.	
Week 5	20/09/2025 – 30/09/2025	Studied preprocessing techniques for camera input and embedding stability.	Studied offline speech recognition frameworks and wake-word strategies.	Researched SGDClassifier and partial_fit for incremental learning.	Finalized technology stack and justified edge-first design.	
Week 6	01/10/2025 – 10/10/2025	Designed module-level responsibilities for vision components.	Designed GPIO-based feedback mechanisms (LED / relay).	Designed confidence thresholding and correction handling logic.	Structured backend responsibilities and API flow.	
Week 7	11/10/2025 – 20/10/2025	Prepared detailed explanation of feature embeddings for report and presentation.	Designed user confirmation dialogue flow.	Finalized learning vs inference mode separation.	Integrated all module flows into a single coherent system diagram.	
Week 8	21/10/2025 – 31/10/2025	Reviewed vision-related literature for report writing.	Assisted in refining human-in-the-loop explanation sections.	Drafted learning module explanation for documentation.	Drafted system architecture and algorithm sections of the report.	
Week 9	01/11/2025 – 10/11/2025	Verified vision module descriptions in presentation slides.	Verified interaction and feedback slides.	Verified learning and knowledge storage slides.	Verified UML, workflow, and algorithm flowcharts.	
Week 10	11/11/2025 – 20/11/2025	Participated in slide review and consistency checking.	Participated in slide review and consistency checking.	Participated in slide review and consistency checking.	Consolidated Phase-I slides and ensured alignment with evaluation criteria.	
Week 11	21/11/2025 – 05/12/2025	Assisted in refining module boundaries.	Assisted in refining interaction flow for future implementation.	Assisted in refining learning logic descriptions.	Updated results, work status, and conclusion slides to reflect Phase-I completion.	
Week 12	06/12/2025 – 18/12/2025	Final verification of vision module responsibilities and Phase-II planning.	Final verification of interaction responsibilities and Phase-II planning.	Final verification of learning module responsibilities and Phase-II planning.	Finalized presentation, contribution mapping, and Phase-I completion documentation.	
Week 13	20/12/2025 – 31/12/2025	Started implementing vision embedding pipeline using MobileNet/MediaPipe.	Began setting up Vosk speech recognition environment.	Started designing prototype storage format and update logic.	Set up FastAPI backend structure and initial routes.	
Week 14	01/01/2026 – 10/01/2026	Integrated camera input with embedding extraction.	Implemented basic voice-to-text pipeline using Vosk.	Implemented initial prototype store (JSON-based).	Connected frontend camera with backend perceive API.	
Week 15	11/01/2026 – 20/01/2026	Tested embedding stability across frames.	Improved voice command parsing and confirmation flow.	Implemented similarity computation and threshold logic.	Built basic HTML/JS UI for learn and infer flow.	
Week 16	21/01/2026 – 30/01/2026	Assisted in debugging embedding dimension and format issues.	Integrated voice input with learning workflow.	Added multi-prototype learning and reinforcement logic.	Integrated learn, infer, and confirm APIs with frontend.	
Week 17	31/01/2026 – 02/02/2026	Helped tune similarity thresholds and confidence logic.	Assisted in debugging speech failures and fallback logic.	Implemented drift handling and prototype pruning.	Debugged CORS, API errors, prototype storage, and UI refresh issues.	

Meeting Date	Review of the task completion	Meetup summary	New task assigned	Expected time taken for the assigned task	Signature